

LAB EXPERIMENT RECORD

SET 7 — Customer Demographics Analysis

Department	AIDS
Course Code / Course Name	DSA0602 / Data Handling and Visualization
Experiment	Data Visualization using R
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Date	3 October 2026
Day	Saturday

A. Aim

To analyze customer demographic information using **bar charts, pie charts, tables, and an interactive Tableau dashboard.**

B. Dataset

Customer ID	Age	Gender	Income (\$)
1	28	Female	50000
2	35	Male	60000
3	42	Female	75000

Program 1 — Bar Chart of Customer Ages

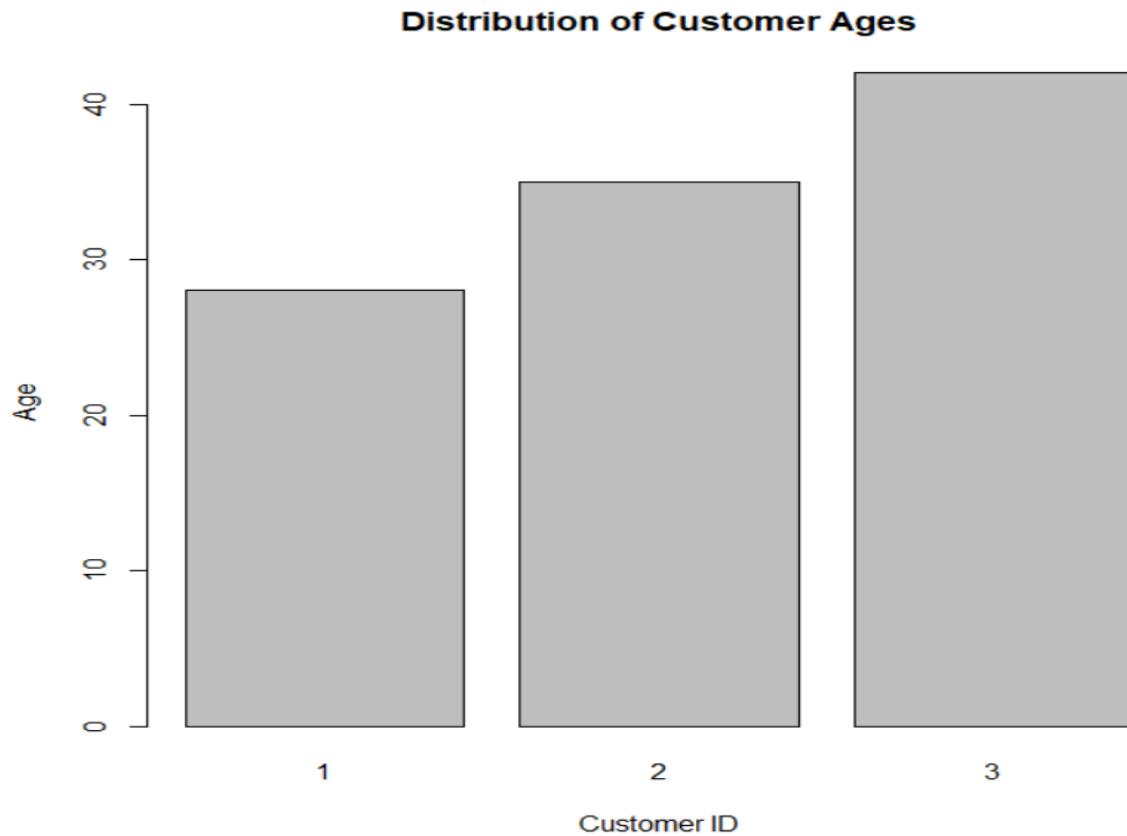
```
customer_id <- c(1, 2, 3)
```

```
age <- c(28, 35, 42)
```

```
customer <- data.frame(customer_id, age)
```

```
barplot(  
  customer$age,  
  names.arg = customer$customer_id,  
  main = "Distribution of Customer Ages",  
  xlab = "Customer ID",  
  ylab = "Age"  
)
```

```
text(  
  seq_along(customer$age),  
  customer$age,  
  labels = customer$age,  
  pos = 3  
)
```



Observation

Customer 3 has the highest age of **42 years**, while Customer 1 has the lowest age of **28 years**.

Program 2 — Pie Chart of Customer Gender

```
gender <- c("Female", "Male", "Female")
```

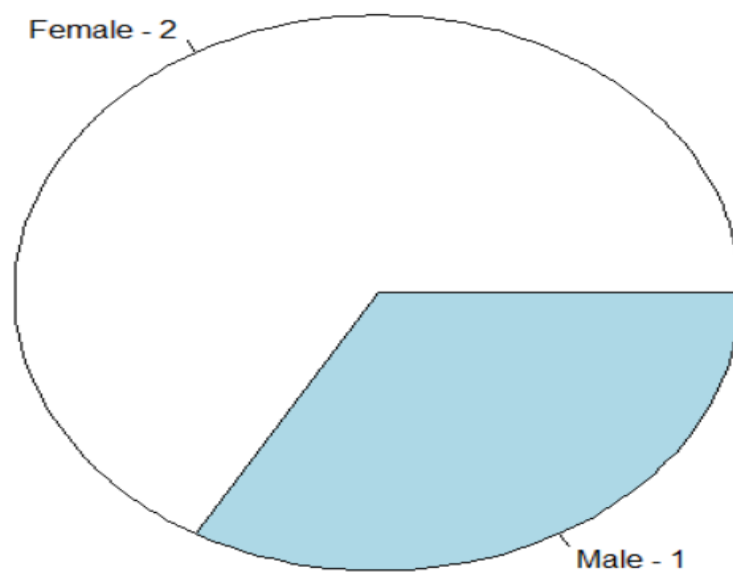
```
gender_count <- table(gender)
```

```
pie(  
  gender_count,  
  main = "Customer Distribution by Gender",  
  labels = paste(  

```

```
names(gender_count),  
gender_count,  
sep = " - "  
)  
)
```

Customer Distribution by Gender



Observation

There are **2 female customers** and **1 male customer** in the given dataset.

Program 3 — Customer Demographic Table

```
customer_data <- data.frame(  
  Customer_ID = c(1, 2, 3),  
  Age = c(28, 35, 42),
```

```
Gender = c("Female", "Male", "Female"),  
Income = c(50000, 60000, 75000)  
)
```

```
print(customer_data)
```

```
>  
> print(customer_data)  
  Customer_ID Age Gender Income  
1           1  28 Female  50000  
2           2  35   Male  60000  
3           3  42 Female  75000  
>  
>
```

Observation

The table displays the **Customer ID**, **Age**, **Gender**, and **Income** of each customer.

Program 4 — Tableau Dashboard

Bar Chart

1. Import the dataset into Tableau.
2. Drag **Customer ID** → Columns.
3. Drag **Age** → Rows.
4. Select **Bar Chart**.
5. Enable **Show Mark Labels**.
6. Title it **Customer Age Distribution**.

Pie Chart

1. Create a new worksheet.
2. Select **Pie Chart**.

3. Drag **Gender** → Color.
4. Drag **Customer ID** → Angle.
5. Title it **Customer Gender Distribution**.

Table

1. Create another worksheet.
2. Drag **Customer ID, Age, Gender, Income** into the view.
3. Select **Text Table**.

Dashboard

1. Select **New Dashboard**.
2. Add the bar chart.
3. Add the pie chart.
4. Add the demographic table.
5. Add **Gender** and **Age** as filters if required.
6. Title the dashboard **Customer Demographics Analysis**.

Customer Demographics Analysis

Select Gender:

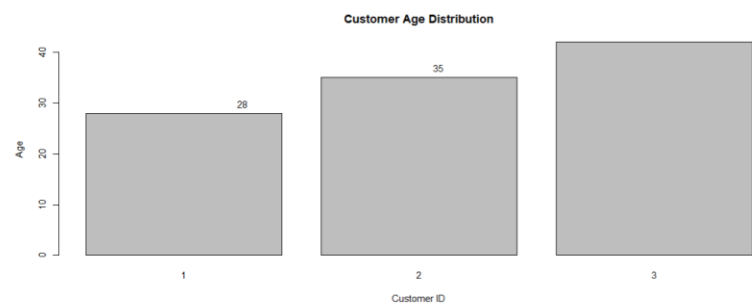
All

Select Age Range:

20 50

20 23 26 29 32 35 38 41 44 47 50

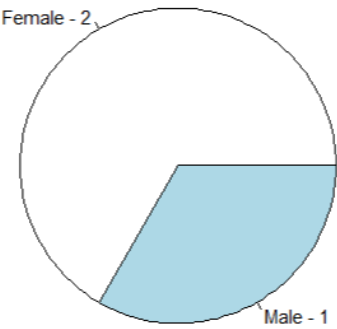
Customer Age Distribution



Customer Gender Distribution

7.

Customer Gender Distribution



Customer Demographic Table

Customer_ID	Age	Gender	Income
1.00	28.00	Female	50000.00
2.00	35.00	Male	60000.00
3.00	42.00	Female	75000.00

8. _____

C. Overall Procedure

- 1. Create the customer demographic dataset.
- 2. Visualize customer ages using a bar chart.
- 3. Represent gender distribution using a pie chart.
- 4. Display demographic information in a table.
- 5. Combine all visualizations in Tableau.
- 6. Add filters for interactive exploration.

D. Result

Customer demographic information was successfully analyzed using **bar charts, pie charts, tables, and an interactive Tableau dashboard.**

E. Key R Functions Used

Function	Purpose
c()	Create vectors
data.frame()	Create dataset
barplot()	Create bar chart
pie()	Create pie chart
table()	Count categories
text()	Add labels
print()	Display table